

Inside Test



Exercise (Solution)

Inside Test

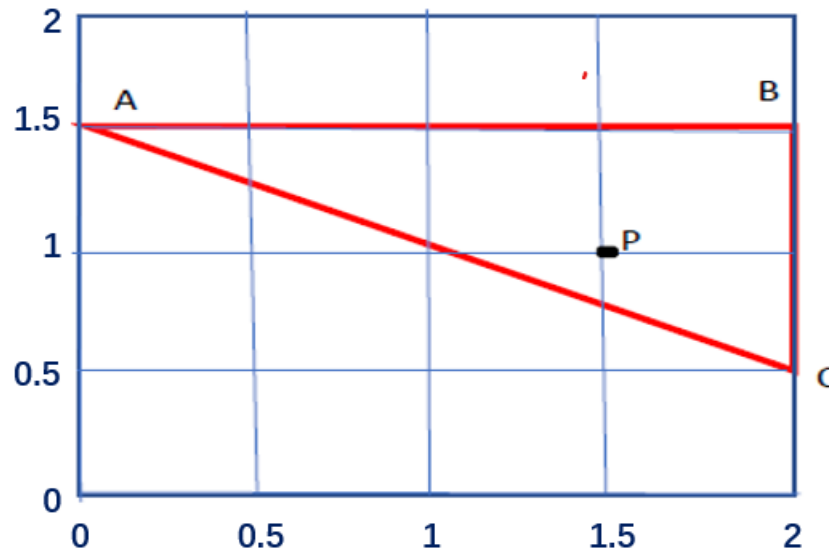
- ◆ Compute the Barycentric Coordinates α , β and γ for the following triangle ABC, and point P. Is the point inside the triangle?

A(0, 1.5)

B(2, 1.5)

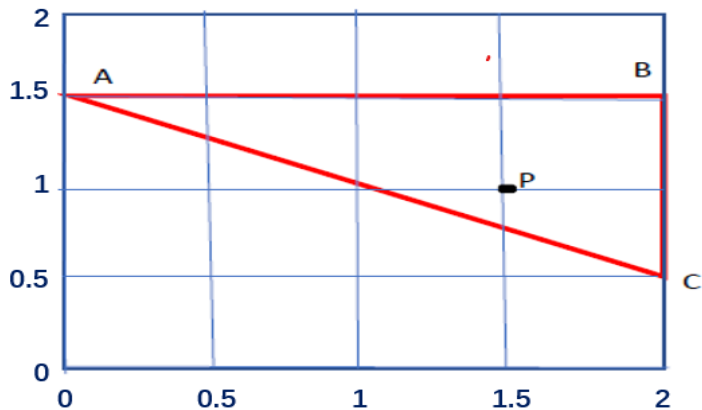
C(2, 0.5)

P(1.5, 1)



Calculation of α

$$\text{line}(F, S, T) = (S_y - F_y)T_x + (F_x - S_x)T_y + S_x F_y - F_x S_y$$



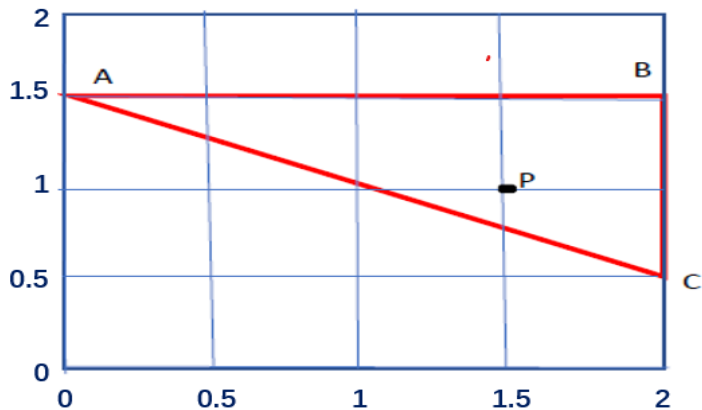
$$\begin{aligned}\text{line (B,C,P)} &= (0.5-1.5)1.5 + (2-2)1 + [(2 \times 1.5) - (2 \times 0.5)] \\ &= 0.5\end{aligned}$$

$$\begin{aligned}\text{line (B,C,A)} &= (0.5-1.5)0 + (2-2)1.5 + [(2 \times 1.5) - (2 \times 0.5)] \\ &= 2\end{aligned}$$

$$\begin{aligned}\alpha &= \text{line (B,C,P)} / \text{line (B,C, A)} \\ &= 0.5 / 2 \\ &= 0.25\end{aligned}$$

Calculation of β

$$\text{line}(F, S, T) = (S_y - F_y)T_x + (F_x - S_x)T_y + S_x F_y - F_x S_y$$



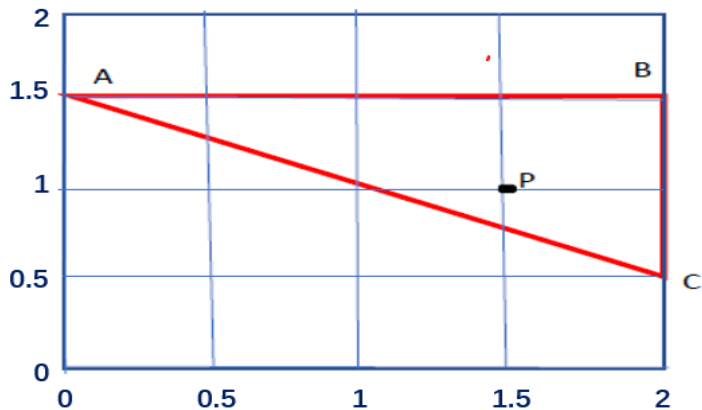
$$\begin{aligned}\text{line (A,C,P)} &= (0.5-1.5)1.5 + (0-2)1 + [(2 \times 1.5)-(0 \times 0.5)] \\ &= -0.5\end{aligned}$$

$$\begin{aligned}\text{line (A,C,B)} &= (0.5-1.5)2 + (0-2)1.5 + [(2 \times 1.5)-(0 \times 0.5)] \\ &= -2\end{aligned}$$

$$\begin{aligned}\beta &= \text{line (A,C,P)} / \text{line (A,C, B)} \\ &= -0.5 / -2 \\ &= 0.25\end{aligned}$$

Calculation of γ

$$\text{line}(F, S, T) = (S_y - F_y)T_x + (F_x - S_x)T_y + S_xF_y - F_xS_y$$



$$\begin{aligned}\text{line (A,B,P)} &= (1.5-1.5)1.5 + (0-2)1 + [(2 \times 1.5)-(0 \times 1.5)] \\ &= 1\end{aligned}$$

$$\begin{aligned}\text{line (A,B,C)} &= (1.5-1.5)0.5 + (0-2)0.5 + [(2 \times 1.5)-(0 \times 1.5)] \\ &= 2\end{aligned}$$

$$\begin{aligned}\gamma &= \text{line (A,B,P)} / \text{line (A,B, C)} \\ &= 1 / 2 \\ &= 0.50\end{aligned}$$

Is the point P inside the triangle?

α , β and γ are all >0 and <1 so the point is algebraically inside the triangle.